PARALEARN RMS PLATFORM

Manual QA Testing Matrix & Feature Validation Checklist

Target School Subdomain: bfa (Bright Future Academy)
App Web URL: http://bfa.localhost:3000 (or https://bfa.pln.ng)
Backend API URL: http://localhost:8000/api/v1
Document Date: September 2026

Admin Credentials: admin@brightfuture.ng / Admin@123 [BFA-ADM-26-0001]
VP Credentials: new_accountant@gmail.com / BFA-VP-26-0001
Bursar Credentials: testaccountant@mail.com / BFA-ACC-26-0001
Teacher / Student: BFA-T-26-0001 / BFA-S-26-0001

1. AUTHENTICATION & MULTI-TENANT ACCESS CONTROL

#	Test Scenario	Testing Steps	Expected Result	Result
1.1	**School Admin Login**	1. Navigate to /auth/signin 2. Select Admin tab 3. Enter admin@brightfuture.ng & Admin@123 4. Click Sign In	User authenticated successfully (200 OK). Redirected to Admin Dashboard (/RMS/dashboard).	☐
1.2	**Bursar / Accountant Login**	1. Select Bursar tab on /auth/signin 2. Enter Code BFA-ACC-26-0001 (or email & password) 3. Submit form	Authenticated; redirected directly to dedicated Finance portal (/RMS/finance). Sensitive admin tabs hidden.	☐
1.3	**Vice Principal (VP) Login**	1. Select VP tab on /auth/signin 2. Enter Code BFA-VP-26-0001 3. Submit form	Authenticated; redirected to Academic Administration and Result Approval views.	☐
1.4	**Teacher Login**	1. Select Teacher tab 2. Enter Code BFA-T-26-0001 3. Submit form	Authenticated; views assigned classes, subjects, CA and score submission sheets.	☐
1.5	**Student Login**	1. Select Student tab 2. Enter Code BFA-S-26-0001 3. Submit form	Authenticated; student portal displays invoices, online fee payment, and published report cards.	☐
1.6	**Tenant School Registration**	1. Go to /auth/register-school 2. Fill school details and choose subdomain 3. Submit registration	Tenant school created with dedicated subdomain. Default roles (admin, vp, accountant, teacher, student) auto-seeded.	☐

2. USER DIRECTORY & UNIQUE CODE GENERATION

#	Test Scenario	Testing Steps	Expected Result	Result
2.1	**Unique Code Format Verification**	1. Log in as Admin 2. Go to /RMS/users 3. Inspect role badges and user codes	Correct prefix formatting displayed: - Admin: BFA-ADM-26-XXXX - VP: BFA-VP-26-XXXX - Bursar: BFA-ACC-26-XXXX - Teacher: BFA-T-26-XXXX - Student: BFA-S-26-XXXX	☐
2.2	**Add User with Role Assignment**	1. Click "+ Add User" 2. Select role (e.g. Bursar or Teacher) 3. Enter name and email 4. Submit	User is created and assigned the selected role. Unique code is generated and visible immediately in table.	☐
2.3	**User Directory Search & Filters**	1. Type student code, teacher code, or name in search box 2. Switch between role filter tabs	Table filters matching records dynamically without console errors or layout issues.	☐

3. FINANCE & BURSARY MANAGEMENT (/RMS/FINANCE)

#	Test Scenario	Testing Steps	Expected Result	Result
3.1	**Bank Settlement Configuration**	1. Go to /RMS/finance/settings 2. Select Nigerian bank from dropdown 3. Input 10-digit NUBAN account number	Account name resolves live via Paystack. Saving persists settlement subaccount configuration.	☐
3.2	**Fee Structures Setup**	1. Go to /RMS/finance/fee-structures 2. Create fee item (Tuition, Development, Sports) 3. Assign to class and term	Fee item saved. Breakdown shows compulsory and optional fees correctly.	☐
3.3	**Individual Invoice Generation**	1. Go to /RMS/finance/invoices 2. Click Generate Invoice -> Individual 3. Type student name or code (BFA-S-26-0002)	Autocomplete dropdown displays matching students. Selecting populates invoice and generates single bill.	☐
3.4	**New Student Invoicing Detection**	1. On invoice generation modal, check New Students notification tab	System automatically identifies newly registered students lacking invoices for active term.	☐
3.5	**Batch Invoicing by Class**	1. Select Batch Generation 2. Choose target class and term 3. Submit batch creation	Invoices generated for all enrolled students in the class with accurate fee calculation.	☐
3.6	**Online Fee Payment (Paystack)**	1. Log in as Student (BFA-S-26-0001) 2. Open unpaid invoice and click Pay Now 3. Complete test checkout	Paystack modal completes transaction. Webhook triggers status update to PAID. Receipt issued.	☐
3.7	**Offline / Manual Payment Entry**	1. As Bursar, select unpaid invoice 2. Click Record Payment (Bank Transfer / Cash / POS) 3. Enter reference & amount	Payment logged; invoice balance updates (PARTIALLY_PAID / PAID). Payment receipt printable.	☐

4. ACADEMIC & REPORT CARD MANAGEMENT (RMS)

#	Test Scenario	Testing Steps	Expected Result	Result
4.1	**Continuous Assessment Scoring**	1. As Teacher, select subject score sheet 2. Input scores for CA1, CA2, Exam 3. Save scores	Total score and grade (A, B, C, D, E, F) computed automatically based on grading scale.	☐
4.2	**Psychomotor Domain Ratings**	1. Open Psychomotor assessment screen 2. Score behavioral and skill traits (scale 1 to 5)	Ratings saved reliably and reflected on student term report card.	☐
4.3	**Teacher & Principal Remarks**	1. Add teacher comments and principal remarks 2. Test automated remark suggestions	Remarks saved and positioned on final report card layout.	☐
4.4	**Digital Signing & Approval**	1. As Principal/VP, open Sign Report Modal 2. Apply digital signature and approve	Digital signature, timestamp, and verification hash applied. Status changes to SIGNED.	☐
4.5	**Report Card Audit Trail**	1. Click Audit Trail on any report card	Modal displays sequential lifecycle timeline (Draft -> Submitted -> Reviewed -> Signed -> Published).	☐
4.6	**Bulk Report Distribution**	1. Open Bulk Share Report Modal 2. Select class and click Dispatch	Generates and dispatches student report cards via PDF download or email notification.	☐

#	Test Scenario	Testing Steps	Expected Result	Result
4.7	**Fee Paywall Guard on Reports**	1. Set fee threshold (e.g. 100% paid required) 2. As student with outstanding balance, view report	Paywall guard blocks result view and displays notice to settle outstanding invoice balance.	☐

5. SECURITY & ROLE PERMISSIONS CHECK

#	Test Scenario	Testing Steps	Expected Result	Result
5.1	**Bursar Route Restriction**	1. Log in as Bursar 2. Attempt direct URL navigation to /RMS/users or /RMS/reports/sign	Access denied or redirected to /RMS/finance. Non-bursary features inaccessible.	☐
5.2	**Teacher Route Restriction**	1. Log in as Teacher 2. Attempt direct navigation to /RMS/finance/settings	Access Forbidden (403). Cannot access financial settlement setup.	☐
5.3	**Student Route Restriction**	1. Log in as Student 2. Attempt navigation to teacher score entry or admin settings	Access Forbidden (403). Restricted strictly to student portal views.	☐

Tester Name: _____ **Lead Reviewer:** _____

QA Tester Signature & Date **Quality Assurance Lead Approval & Date**